“Class of Customer” Question from the US Economic Census

by

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Abstract

The Economic Census (EC) collects detailed information on the class of customers served by establishments—for example, the share of an establishment's sales to other businesses or to government entities—for a subset of sectors in the economy. In this paper, we evaluate the data from the "Class of Customer" question from the EC, with a particular focus on sales to the government. These data have seldom been used in empirical research and are unique in that they enable researchers to link establishment-level Census data with information on government procurement.

We compile and analyze large volumes of publicly available tabulated data about the class of customer question over time. Using these data, we document three main findings. First, total sales to government from establishments covered by the class of customer question account for approximately 4 percent of GDP—just under half of total government procurement as measured in the national accounts. Second, the sectoral distribution of government expenditure is significantly different from that of private sector spending. Certain industries, such as Construction and Professional, Scientific, and Technical Services, account for a much larger share of government expenditure relative to private sector expenditure. Third, sales to the government make up a substantial portion of total sales in several sectors—for instance, 70 percent in Facilities Support Services, 30 percent in Waste Treatment and Disposal, and 17 percent in Construction. Finally, we use the microdata to examine nonresponse rates to the class of customer question across establishments based on the number of employees.

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1 Introduction

The Economic Census (EC) includes a "Class of Customer" question for selected sectors, asking establishments to report the share of sales to different types of customers such as businesses, governments, and households. This question provides a rare, establishment-level view of demand composition that is not available in most other data sources.

For example, the 2017 EC asked establishments in Management Consulting Services (54161) to report the percentage of the establishment's total operating receipts by its class of customer, such as other businesses or the federal government. See Figure 1 for the full class of customer question for Management Consulting Services. The EC publicly releases tabulated responses, allowing researchers to analyze customer composition across sectors. For example, in the case of Management Consulting Services, data from the Census website indicates that this industry has 139,304 establishments with $214.1 billion in total operating receipts (variable RCPTOT or item 5 of the EC). Table 1 reports the breakdown in total operating receipts for this sector based on the public tabulations. For example, the table reports that 10.4 percent of this sector's sales were to the federal government, and 4.7 percent of sales were to state and local governments. The publicly available tabulations also report that the response coverage of the class of customer—defined as the share of total sector sales contributed by establishments that provided usable responses to the question inquiry—was between 50 and 60 percent.

Figure 1. Class of Customer Question for Management Consulting Services (2017 EC)

ITEM 20: CLASS OF CUSTOMER

What was the percentage of this establishment's total operating receipts (reported in item 5) by class of customer?

1. Business firms and farms
2. Not-for-profit organizations
3. Federal government
4. State and local governments
5. Household consumers and individuals

Sum of lines 1 through 5 should equal 100%.

Despite its long-standing inclusion in the EC, the class of customer question has been largely overlooked in empirical research. We believe that this data could be of value to researchers and especially those studying the effects of government procurement on establishments/firms and the economy as a whole. For example, a series of recent papers use detailed data from USAspending—a publicly accessible federal database that tracks government contracts and financial assistance—to study a range of topics, including the composition and timing of federal procurement contracts, government-sponsored R&D, and the effects of fiscal stimulus.[1] In Section 5, we will discuss the advantages and disadvantages of using the class of customer data from the EC and data from USAspending.

Table 1. Sales in Management Consulting Services by Class of Customer, 2017

Sales to	Percentage of Sales
Business firms and farms	74.8
Not-for-profit organizations	5.9
Federal government	10.4
State and local governments	4.7
Household consumers and individuals	4.2

Notes: Table 1 reports the results from publicly tabulated data from the class of customer question in the 2017 EC for Management and Consulting Services (54161). Column 1 reports the customer class, and column 2 reports the fraction of sales that corresponds to that customer class.

This paper provides the first comprehensive assessment of the class of customer question in the EC. We describe its sectoral coverage since the 1972 EC, summarize key patterns in the public tabulations, and compare EC estimates with alternative sources of procurement data. We also use restricted-use microdata to examine nonresponse behavior across establishment size classes as a way of assessing the quality of the data.

The remainder of this working paper is structured as follows. Section 2 describes the sectoral and historical coverage of the class of customer question. Section 3 presents descriptive statistics on sales by customer class across sectors and years. Section 4 compares estimates with national accounts data, and Section 5 contrasts the EC data with other procurement datasets such as USAspending. Section 6 outlines alternative sources for customer class information, and Section 7 uses microdata to assess nonresponse rates by establishment size. Section 8 concludes.

[1]The topics covered by these papers include the composition of government expenditure (Cox et al. Forthcoming, Liebman and Mahoney 2017, Decarolis et al. 2020), government sponsored R&D (Bruce, de Figueiredo and Silverman 2019 and Belenzon and Cioaca 2021), effects of fiscal policy (Auerbach, Gorodnichenko and Murphy 2019, Auerbach, Gorodnichenko and Murphy 2020, Auerbach, Gorodnichenko and Murphy 2022, and Barattieri, Cacciatore and Traum 2023).

2 Sectoral Coverage

2.1 Overview of Sectoral Coverage Using Publicly Tabulated Data

This subsection documents which sectors are covered by the class of customer question in the EC, based on publicly available tabulations from 2002 to 2017.[2]

Table 2. Summary of Customer Class Information (2002–2017)

NAICS code	Sector Name	Customer Class
22	Utilities	2002, 2007, 2012, 2017
23	Construction	2002^{+}, 2007^{+}, 2012^{+}, 2017^{+}
42	Wholesale trade	2002, 2007, 2012, 2017
44–45	Retail trade	2002, 2007, 2012, 2017
51	Information	2002*, 2007*, 2012*, 2017*
53	Real estate and rental and leasing	2002*, 2007*, 2012*, 2017*
54	Professional, scientific, and technical services	2002*, 2007*, 2012*, 2017*
56	Administrative and support and waste management and remediation services	2002*, 2007*, 2012*, 2017*
72	Accommodation and food services	2007, 2012, 2017
81	Other services (except public administration)	2002*, 2007*, 2012*, 2017*

Notes: Table 2 reports the sectors and years for which publicly tabulated data are available for the class of customer question for at least one four-digit industry group in that sector. The data used in the analysis was downloaded from the Census website. Column 1 reports the two-digit NAICS code for the sector. Column 2 reports the sector name. Column 3 indicates the years in which at least one four-digit industry group was asked the class of customer question in the Economic Census.

$^{+}$ Customer class information is from the Census of Construction's question on project ownership (see Section 2.2 for more details).

* Not all the industry groups (four-digit NAICS) within the sector (two-digit NAICS) have customer class information. See Table A1 for details.

Table 2 includes a summary of the two-digit NAICS codes for which class of customer information is available at least for one four-digit industry group in the ECs during 2002–2017 in the publicly tabulated data. We find that the class of customer question was asked consistently and across all industries within the Utilities (22), Wholesale Trade (42), and Retail Trade (44–45) sectors during this period. As discussed further in Section 2.2, the Construction sector (23) was asked a similar question on project ownership, and we include this sector in Table 2 accordingly. We find that the construction sector has also been consistently asked this question during this period (i.e., 2002–2017). In addition to sectors with consistent coverage, sectors such as Accommodation and Food Services (72) began reporting this information in 2007. Other sectors—including Information (51), Real Estate and Rental

[2]It is interesting to note that, while not all establishments are asked the class of customer question, in the past there have been calls to increase the sectors covered by the class of customer question. Haltiwanger and Jarmin (2000) suggested expanding the class of customer question to all sectors of the economy, especially to improve the measurement of e-commerce. They wrote: "To get a handle on the prevalence and magnitude of e-commerce, we suggest that the Census Bureau include class of customer and method of selling questions on all Economic Censuses and Annual Surveys."

and Leasing (53), and Professional, Scientific, and Technical Services (54)—contain partial coverage, with the class of customer question included for only a subset of four-digit industry groups.

Table A1 reports similar information but at the four-digit NAICS level. The table indicates whether class of customer information was available for the 2002, 2007, 2012, and 2017 EC. An industry is marked as having coverage in a given year if customer class information is reported for that industry at the aggregate level. This table was constructed using publicly available data. The table highlights one area of interest for data users, which is that users should be aware of the change in coverage even in the four-digit NAICS industries. These changes highlight the need for caution when conducting comparisons across time with this data.

In sum, while certain sectors are consistently covered by the class of customer question in the EC, many others reflect changes in coverage, underscoring the importance of closely tracking sectoral scope when using these data for empirical research.

2.2 Construction

While the Construction sector (NAICS 23) is not asked the standard class of customer question (item 20) in the EC, similar information is collected through a separate item in the Census of Construction. Specifically, the 2017 EC includes a Special Inquiries (SI) item that asks establishments to report the share of construction activity by ownership type (see Figure 2). The response categories—private businesses and individuals, state and local governments, and federal government—mirror the structure of the class of customer question used in other sectors. Throughout this paper, we incorporate Construction sector statistics using this alternative measure, which serves as a close proxy for customer class.

2.3 Sectors Excluded from the Class of Customer Question (2017 EC)

Table 3 contains the list of sectors for which no class of customer information is available in the 2017 EC. These sectors include Mining, Manufacturing, Transportation and Warehousing, and service sectors such as Finance and Insurance. Researchers should be aware of these sectoral limitations if they choose to use this data for their research.

2.4 Historical Analysis of Sectoral Coverage

In this subsection, we describe a historical analysis conducted by the authors to determine the sectoral coverage of the class of customer question going back to the 1972 EC. To do so,

Figure 2. Ownership of Projects Question from the 2017 Economic Census of Construction

ITEM 28: SPECIAL INQUIRIES - OWNERSHIP OF PROJECTS

Percent of the construction work reported in item 5 (Sales, Shipments, Receipts, or Revenue) on projects owned by the following:

1. Private businesses and individuals
2. State and local governments
3. Federal Government

(Percentages reported in lines 1–3 should equal 100%.)

Table 3. Sectors in Which EC Does Not Ask for Class of Customer (2017)

Sector	Sector name
11	Agriculture, Forestry, Fishing and Hunting
21	Mining, Quarrying, and Oil and Gas Extraction
31–33	Manufacturing
48–49	Transportation and Warehousing
52	Finance and Insurance
55	Management of Companies and Enterprises
61	Educational Services
62	Health Care and Social Assistance
71	Arts, Entertainment, and Recreation
92	Public Administration

Notes: Table 3 reports the sectors that were not asked the class of customer question in the 2017 EC. Column 1 reports the two-digit NAICS code of the sector. Column 2 reports the name of the sector. Note that Census of Construction Industries asks about the ownership of projects and gives information similar to the class of customer question (see Section 2.2 for more details).

we conducted a historical analysis of the documentation issued for past ECs to determine whether the class of customer question was asked in relevant sectors.[3]

2.4.1 Sectoral Coverage through Time

Table 4 summarizes the coverage of the class of customer question by economic area, including the years in which the question was asked. These documents indicate that the sectors that have been traditionally asked the class of customer question are the Wholesale Trade and Retail Trade sectors. For example, the class of customer question has been included in the Retail and Wholesale Trade sectors since the 1972 EC. This analysis also indicates that the 1987 EC was the first instance in which the class of customer question was asked for some service sectors.[4] We find that the 1992 EC was the first instance in which the class of customer question was asked for utilities.

Table 4. Class of Customer Question Coverage by Economic Area

Economic area	Years covered	Industries covered
Census of Retail Trade	Starting 1972	All industries
Census of Wholesale Trade	Starting 1972	All industries
Census of Construction Industries	Starting 1972[+]	All industries[+]
Census of Manufactures	1977 and 1987	All industries
Census of Services	Starting 1987	Some industries
Census of Transportation, Communications, and Utilities	Starting 1992	Some industries
Census of Finance, Insurance, and Real Estate	None	None
Census of Mining	None	None

Notes: Table 4 describes the coverage of the class of customer by economic area. Column 1 indicates the economic area. Column 2 indicates the years covered by the class of customer question. We include a particular year if at least one industry is covered by the class of customer question. Column 3 describes the industry coverage (e.g., whether all or only some industries are covered).

[+] Customer class information is from the Census of Construction's question on project ownership (see Section 2.2 for more details).

Our analysis of historical Census Bureau reports indicates that data relating to the ownership of projects have been collected in the Census of Construction Industries since at least 1972.[5] As discussed in Section 2.2, the Census of Construction Industries has included

[3]The documents we reviewed include the documentation issued by the Census Bureau relating to the EC of 1967, 1972, 1977, 1982, 1987, and 1992. These documents include the following sources: Bureau of the Census (1972), Bureau of the Census (1976), Bureau of the Census (1982), Bureau of the Census (1984), Bureau of the Census (1990*b*), Bureau of the Census (1996), US Census Bureau (2000), US Census Bureau (2006), and US Census Bureau (2011).

[4]See Bureau of the Census (1991*b*) for tabulated service statistics.

[5]Our historical analysis finds that the Census has collected information relating to ownership of problems for the ECs since 1972. The following are the sources used in this analysis for each of the EC years: 1972 (Bureau of the Census 1975), 1977 (Bureau of the Census 1979), 1982 (Bureau of the Census 1985), 1987 (Bureau of the Census 1991*a*), 1992 (Bureau of the Census 1995), 1997 (Bureau of the Census 2000).

a project ownership question since at least 1972, which we treat as a close substitute for customer class.

These patterns underscore the evolution of the EC and the need for researchers to consider historical changes in sectoral coverage when working with these data.

2.4.2 Historical coverage of manufacturing

Table 3 shows that establishments in the Manufacturing sector are not currently asked the class of customer question. In the historical analysis, however, we find that the class of customer question has been asked in the Manufacturing sector in the past. For example, the class of customer question was asked in 1967, 1977, and 1987 (as opposed to every five years for the Wholesale Trade and Retail Trade sectors).[6] The EC did not ask the class of customer question for the Manufacturing sector after the 1987 EC.[7]

Given that the last available information about the class of customers for manufacturing establishments is the 1987 EC, it is helpful to study these statistics to better understand the distribution of the class of customer question for these establishments. We reproduce the published statistics from the 1987 Census of Manufactures in Table 5 (see Bureau of the Census 1992). The table indicates that most of the shipments from manufacturing establishments were sent to other companies.[8] For example, shipments to other establishments of the same company account for 20 percent of total manufacturing shipments. Shipments to all other customers, such as wholesalers, retailers, and manufacturers, account for 73 percent of total shipments. Finally, 7 percent of shipments are to federal, state, and local governments, and 11 percent are listed as all others.

[6]The Bureau of the Census (1990*b*), which is the documentation for the 1987 EC, indicates: "Distribution of Sales by Class of Customer provides information on the primary channels through which manufacturers distributed their products, for example, through retailers, wholesalers, or other manufacturers. Shipments to other establishments within the same company are tallied separately. The data are based on a survey of companies conducted as part of the Census of Manufactures. Comparable data were last published for 1977."

[7]In addition, the Census Bureau collected information about manufacturing establishment sales to the federal government in the 1987 and 1992 EC as part of the Manufacturers' Shipments to Federal Government Agencies Report. Bureau of the Census (1992), which is the documentation related to the 1987 EC, describes this survey as follows: "Manufacturers' Shipments to Federal Government shows estimates from a special survey on the value of manufactured goods shipped to the Department of Defense, the Department of Energy, the National Aeronautics and Space Administration, and all other Federal agencies, for 69 four digit industries that tend to have large Federal contacts."

[8]Bureau of the Census (1992) discusses this finding in the introduction. This document indicates that: "Manufacturing production is usually carried on for the wholesale market, for transfers to other plants of the same company, or to the order of industrial users rather than for direct sale to the household consumer. Some manufacturers in a few industries sell chiefly at retail to household consumers through the mail, through house-to-house routes, or through salespersons. Some activities of a service nature (enameling, engraving, etc.) are included in manufacturing when they are performed primarily for trade. They are considered non-manufacturing when they are performed primarily to the order of the household consumer."

Using publicly tabulated data from the 1987 EC, we can also compare the importance of sales of manufacturing establishments to the government relative to the wholesale sector. This exercise would give us some information about the importance of the omission of the manufacturing sector from the class of customer question if the interest is studying government procurement. We use data from Table 5, along with information from the Bureau of the Census (1990*a*) that has results from the Census of Wholesale Trade. We find that manufacturing establishments sold approximately $157.3 billion to the government (federal, state, and local), whereas the Wholesale sector sold approximately $63.1 billion.[9] Thus, sales from manufacturing establishments to the government were more than twice the sales from wholesale establishments in 1987.

Table 5. Manufacturers' Shipments by Product Group and Class of Customer: 1987

		Value (millions)	Percent
	Total product shipments	2,349,599.7	100
Shipments to other establishments of the same company	Wholesale establishments[a]	225,130.4	10
	Retail store and outlets	30,911.2	1
	Other manufacturing establishments	207,276.4	9
	Other nonmanufacturing establishments	11,451.9	(Z)[b]
Shipments to all other customers	Wholesalers	502,356.1	21
	Retailers	380,356.3	16
	Manufacturers	576,027.7	25
	Federal, state, and local governments	157,284.2	7
	All others	258,805.6	11

Notes: Table 5 reports the publicly tabulated information regarding the class of customer question in the 1987 Census of Manufactures, which was the last EC to ask the class of customer question in manufacturing. Column 1 indicates whether the information corresponds to shipments to other establishments within the firm or to other firms. Column 2 reports the destination of the shipments. Column 3 reports the value of product shipments in millions. Column 4 reports the percentage of total product shipments accounted for by each row.
[a] Shipments to wholesale establishments of the same company included sales offices.
[b] (Z) indicates "Less than half the unit shown."
Source: Table 1 in Bureau of the Census (1992).

This finding suggests that manufacturing establishments are an important source of sales to the government. Thus, researchers who seek a more complete view of sales of *goods* to the government could focus their work on historical data when the Census of Manufactures asked the class of customer question. Alternatively, researchers could supplement the more recent data from the class of customer question with additional sources, which we will discuss further in Sections 5 and 6.

[9]Table 1 of Bureau of the Census (1990*a*), "Sales by Class of Customer for the United States," indicates that the wholesale trade sector had sales of $2,524,726,802,000 and that 2.5 percent of these sales were to federal, state, and local governments.

3 Descriptive Statistics of Publicly Tabulated Data

3.1 Aggregate Statistics

We begin our analysis of the publicly available data by documenting aggregate patterns in total establishment sales and sales to the government across EC years. Table 6 summarizes these patterns and relates them to aggregate output measures. The table reports aggregate statistics from the tabulated data for EC years 2002, 2007, 2012, and 2017. Column 2 reports total sales for all industries with customer class data as a fraction of GDP for that particular year. GDP statistics are from the Bureau of Economic Analysis (BEA).

Table 6. Sales to Government (% of Output)

Year	Sales/GDP (%)	Sales to Gov/GDP (%)	Sales/GO (%)	GDP/GO (%)
2002	97.76	4.99	55.73	57.01
2007	105.85	4.25	58.38	55.15
2012	104.03	4.20	57.86	55.62
2017	98.66	3.98	55.80	56.56

Notes: Table 6 presents the sales to the government as a fraction of output. These figures are based on calculations derived from the publicly tabulated data from the class of customer question at the NAICS four-digits level. Column 1 reports the year. Column 2 reports the total sales across all four-digit industry groups with customer class information as a fraction of GDP. Note that these statistics may be greater than 100 percent since total sales may double-count inter-industry transactions (i.e., sales are gross flows, whereas GDP is a value-added measure). Column 3 indicates the total sales to government across all industry groups as a fraction of GDP. To calculate the total sales to the government, for each four-digit industry group, we multiply the fraction of sales to the government and total sales. We sum this statistic over all four-digit industry groups for a given year. Column 4 reports the total sales across all industries with customer class information as a fraction of gross output (GO). GO is defined as the value of the goods and services produced by the nation's economy. Note that GO includes sales to final users (GDP) and sales to other industries (intermediate inputs). Column 5 reports aggregate GDP divided by GO. Note that we include the construction sector in these calculations since the Census of Construction Industries asks about the ownership of projects and gives information similar to the class of customer question (see Section 2.2 for more details).

We find that for all EC years, the ratio is stable and approximately equal to 100 percent. Note that these statistics may be greater than 100 percent since the total sales of establishments may double-count intermediate inputs. The third column reports the sales to the government for all industries with customer class data as a fraction of GDP.[10] We incorporate the Construction sector using the project ownership variable described in Section 2.2, which closely mirrors the class of customer breakdown. We find that the fraction has declined from 4.99 in 2002 to 3.98 in 2017.

[10] We take the following steps to calculate these statistics:

1. For each four-digit industry group, we multiply the fraction of sales to the government as reported in the publicly tabulated statistics and the total sales.
2. We sum this statistic over all four-digit industry groups for a given year.
3. Divide the sum calculated in step 2 by GDP.

3.2 Purchases by the Government versus the Private Sector

This section contrasts the sectoral composition of government purchases versus private sector purchases, highlighting the distinct patterns in procurement demand across industries. These results show that government demand is concentrated in a few sectors—especially Construction and Wholesale trade—whereas private sector demand is more diffuse and oriented toward consumer-facing industries such as Retail. These results, of course, only include the sectors covered by the class of customer question.

Table 7 reports the distribution of purchases by the government across the EC years 2002, 2007, 2012, and 2017 by two-digit sectors.[11] Thus, it represents the fraction of purchases made by the government that are accounted for by a particular sector, which we refer to as the government share. We sort the table in descending order based on this statistic. We find that the sector with the largest government share is Construction, which accounts for almost a half of government purchases. The second and third sectors are Wholesale Trade and Professional, Scientific, and Technical Services, with 19 percent and 18 percent respectively. These three sectors account for over 80 percent of government purchases. We additionally find that Retail Trade is significantly less important than Wholesale Trade in the government share.

The fourth column of Table 7 reports the private share, which is constructed in a manner similar to the government share. We find that the private share is very different from the government share across sectors. For example, Construction is less important in the private share, whereas Wholesale Trade and Retail Trade are more significant in the private share.

Table 8 reports the government share and private share at the NAICS four-digit level across all ECs' years. Given the large number of four-digit industry groups, the table reports the results for the industry groups with the highest government share. Note that Table 8 does not include Construction, as the publicly available statistics are not available at the four-digit level. As before, the table is sorted in descending order in terms of the government share.

We find that the highest government share by four-digit industry group is Computer

[11] We take the following steps to calculate these statistics:

1. For each four-digit industry group, we multiply the fraction of sales to the government and total sales.
2. Within each two-digit sector, we sum the values from 1 across the four-digit industries and years. This gives us the total sales to government for the sector.
3. We sum the values from 1 across all four-digit industries and years. This gives us the overall sales to government.
4. Finally, we divide the sector sum from 2 to the overall sum in 3.

Table 7. Distribution of Purchases of Government vs. Private Sector, by 2-digit NAICS

NAICS2	Sector Name	Share (%) in government purchases	Share (%) in private sector purchases
23	Construction	47.72	8.39
42	Wholesale trade	18.82	45.86
54	Professional, scientific, and technical services	17.90	5.09
56	Administrative and support and waste management and remediation services	7.96	2.56
44	Retail trade	2.00	19.48
22	Utilities	1.68	3.44
72	Accommodation and food services	1.52	3.73
45	Retail trade	1.05	6.35
81	Other services (except public administration)	0.99	0.93
53	Real estate and rental and leasing	0.38	0.68
51	Information	0.00	3.47

Notes: Table 7 reports the distribution of purchases by government and private sector across two-digit NAICS sectors based on calculations derived from the publicly tabulated data from the class of customer question at the four-digit NAICS level. Column 1 reports the two-digit NAICS code for the sector. Column 2 indicates the sector name. Column 3 represents the sales to the government of that sector across the years over the total sales to the government across all sectors and years. To calculate the total sales to the government from a sector, we multiply the fraction of sales to the government and total sales for each four-digit industry group. We sum this statistic over all four-digit industry groups in a two-digit sector. Then, we divide this sum by the overall total sales to government across the industries and years. See footnote 11 for more details. Column 4 reports the sales to non-government customers of that sector divided across the EC years 2002, 2007, 2012, and 2017 by the total sales to non-government customers across all sectors and years. The table is sorted in descending order by the government share column. Note that the government share column (column 3) sums to 100 percent and similarly for the private share column. In addition, these calculations exclude purchases made by the government or the private sector from sectors that lack class of customer information (e.g., the Manufacturing sector). Note also that we include the construction sector in these calculations since the Census of Construction Industries asks about the ownership of projects and gives information similar to the class of customer question (see Section 2.2 for more details).

Table 8. Distribution of Purchases of Government vs. Private Sector, 4-digit NAICS with Highest Government Share (Excluding Construction)

NAICS4	Industry Name	Share (%) in government purchases	Share (%) in private sector purchases
5415	Computer systems design and related services	21.99	1.62
4234	Professional and commercial equipment and supplies merchant wholesalers	9.62	3.40
5417	Scientific research and development services	6.33	0.40
4236	Household appliances and electrical and electronic goods merchant wholesalers	4.61	3.52
4238	Machinery, equipment, and supplies merchant wholesalers	4.32	3.22
5612	Facilities support services	3.79	0.07
2211	Electric power generation, transmission and distribution	2.63	2.98
4247	Petroleum and petroleum products merchant wholesalers	2.54	6.52
4231	Motor vehicle and motor vehicle parts and supplies merchant wholesalers	2.33	5.23
4244	Grocery and related product merchant wholesalers	2.25	5.41

Notes: Table 8 reports the distribution of purchases from industry groups (four-digit NAICS) with the highest government share. Column 1 reports the NAICS four-digit industry group. Column 2 reports the name of the industry group. Column 3 represents the sales to the government of that industry group over the total sales to the government across all industry groups. To calculate the total sales to the government, we multiply the fraction of sales to the government and total sales for each four-digit industry group. We sum this statistic over all years for each four-digit industry group. Then, we divide this sum by the overall total sales to government across the industries and years. See footnote 11 for more details. Column 4 represents the sales to non-government customers of that sector over the total sales to non-government customers across all sectors. The table is sorted in descending order by the government share column (third column). Note that the Construction sector is not included in these calculations because publicly tabulated data is not available at the four-digit level. In addition, these calculations exclude purchases made by the government or the private sector from sectors that lack class of customer information (e.g., the Manufacturing sector).

Systems Design and Related Services (NAICS 5415). Examples of industries in this sector include Custom Computer Programming Services (541511), Computer Systems Design Services (541512), and Computer Facilities Management Services (541513). As before, the government and private shares are different. For instance, NAICS 5415 has a high government share (21.99 percent) but a much lower private share (1.62 percent), reflecting the government's stronger relative demand for IT services.

3.3 Share of Sales to the Government by Sector

This section uses the customer class information to look at the share of sales that are accounted for by the government across different sectors (e.g., the fraction of Wholesale Trade sales accounted for by the government). The goal is to have a sense of how important government demand is across different sectors of the economy.

Figure 3 shows the top 10 two-digit NAICS sectors in terms of the percentage of sales to the government. The sector for which government demand is most important is Construction (23), with a share of sales to the government between 17 and 25% depending on the year. Other sectors with a high share of sales to the government are Professional, Scientific, and Technical Services (54) with a share of sales to the government between 13 and 16%, and Administrative and Support, and Waste Management and Remediation Services (56) with shares around 12%.

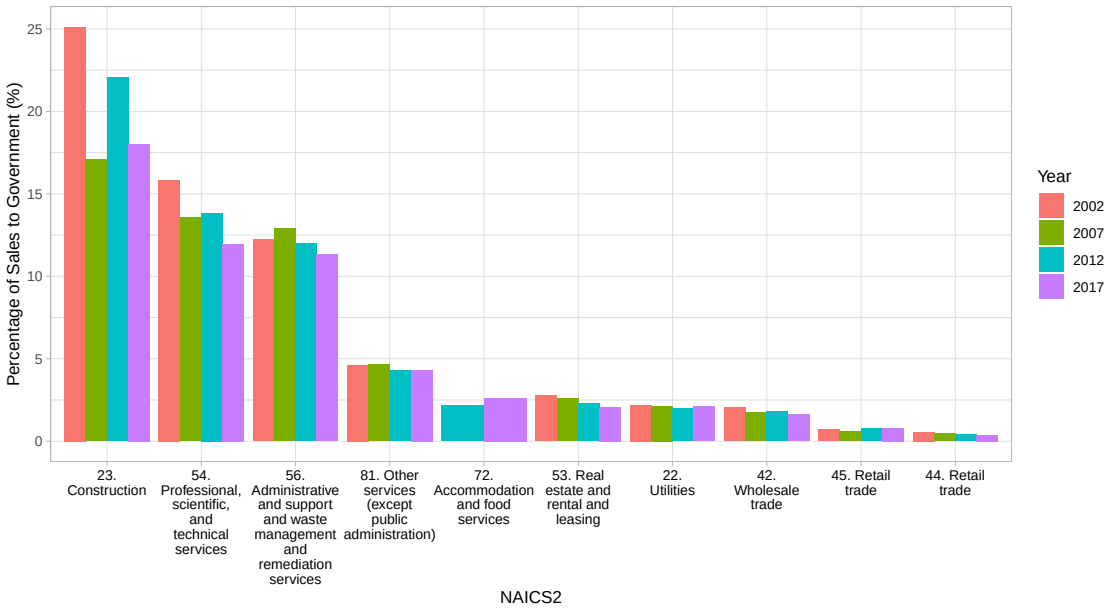
Figure 4 shows similar statistics but at the 4-digit NAICS level. The bars are sorted in a descending order by the sales to government in year 2017. The sectors with the highest share of sales to the government are Facilities Support Services (5612) with shares between 50 and 64%, Waste Treatment and Disposal (5622) with shares around 30%, and Computer Systems Design and Related Services (5415) with shares between 20 and 28%. We exclude Construction from this analysis, as the publicly available statistics are not available at the four-digit level, as discussed in Section 3.2.

3.4 Response Coverage

The EC reports the response coverage of the class of customer question—defined as the share of total sector sales contributed by establishments that provided usable responses to the question.[12] Table A2 summarizes response coverage by sector for the 2002, 2007, and 2012

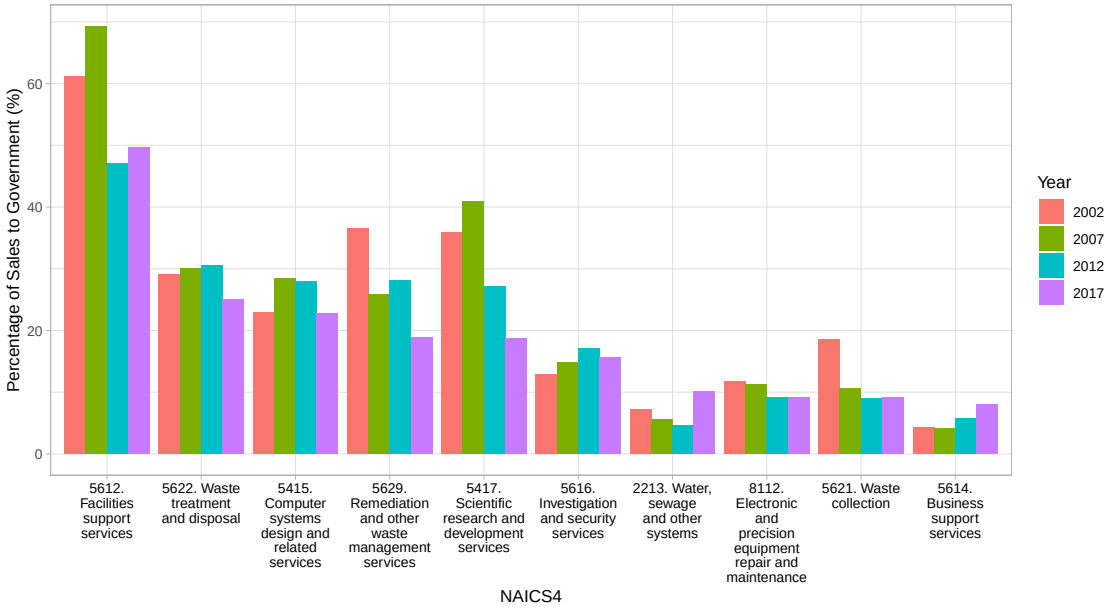
[12]The glossary of fields and variables for the 2017 EC defines the response coverage for the class of customer question as follows: "Sales, value of shipments, or revenue of establishments responding to class of customer inquiry as percent of total sales, value of shipments, or revenue." See https://www.census.gov/programs-surveys/economic-census/year/2017/economic-census-2017/about/fieldsandvariables.html for more information. Note that the computation of the response coverage uses imputed sales for establishments that do not report their sales to Census. For example, the denominator

Figure 3. Top 10 NAICS Sectors (two-digit NAICS): Percentage of Sales to Government



Notes: Figure 3 shows the percentage of sales to government by each sector (two-digit NAICS) for 2002, 2007, 2012, and 2017. To calculate the total sales to the government, we multiply the fraction of sales to the government and total sales for each four-digit industry group. We sum this statistic over all four-digit industry groups in a two-digit sector. We then divide this sum by the total sales of the sector in that year. Note that sector 72 has only two columns because customer class information was provided separately for government sales only in the years 2012 and 2017. Note also that we include the construction sector in these calculations since the Census of Construction Industries asks about the ownership of projects and gives information similar to the class of customer question (see Section 2.2 for more details).

Figure 4. Top 10 NAICS Industry Groups (four-digit NAICS): Percentage of Sales to Government (Sector 23 Construction not included)



Notes: Figure 4 shows the percentage of the sales to the government by industry group (four-digit NAICS) for 2002, 2007, 2012, and 2017. The statistic is defined as total sales to the government in a four-digit industry divided by that industry's total sales. Note also that we exclude the construction sector in these calculations since the Census of Construction Industries data is not available at the four-digits levels (see Section 2.2 for more details).

ECs. We find substantial variation across sectors: Utilities (NAICS 22) and Real Estate and Rental and Leasing (NAICS 53) consistently exhibit higher response coverage rates (typically above 75 percent), whereas Information (NAICS 51) has the lowest coverage, often in the 50–55 percent range. Note that, because the 2017 EC only reports response coverage in intervals (e.g., 50–60 percent), we exclude it from Table A2 and focus on earlier years with exact data.

Finally, the average response coverage across sectors has remained stable at around 70–75 percent from 2002 to 2012 (see the "Mean" column of Table A2).

4 Comparison of US Census from Customer Class versus NIPA Tables

This section compares total establishment sales to the government using the customer class information and total government purchases computed using the National Income and Prod-

of the computation relies on imputed sales for non-responding establishments in order to calculate the total industry sales. Thus, the reliability of this measure depends on the accuracy of the imputation algorithms used by Census.

uct Accounts (NIPA) tables.[13] We define total government purchases from NIPA tables as:

$$\begin{aligned} \text{Government Purchases in NIPA} &= \text{Intermediate Goods and Services Purchased} \\ &+ \text{Gross Investment (Structures)} \\ &+ \text{Gross Investment (Equipment)} \end{aligned} \tag{1}$$

The largest item in this equation is "Intermediate Goods and Services Purchases." For example, in the year 2012, it was approximately $1,000 billion, of which $635 billion were accounted for by services, $297 billion were accounted for by nondurable goods, and $72 billion were accounted for by durable goods. In the same year, gross investment in structures and equipment amounted to around $287 billion and $143 billion, respectively.

Figure 5 shows the evolution of government purchases using the class of customer information and using the NIPA tables for the EC years, both divided by the GDP of each specific year.[14] The main finding from this figure is that government purchases from the customer class variable account for 40–60% percent of total government purchases measured in the NIPA tables. This finding is not surprising given that the customer class information is only available for a subset of sectors in the economy, as discussed in Section 2.

It is worth noting that BEA is an active user of the tabulations derived from the class of customer in the construction of the NIPA, especially with regards to refining the estimates derived from the commodity-flow method. Bureau of Economic Analysis (2006) indicates the following:

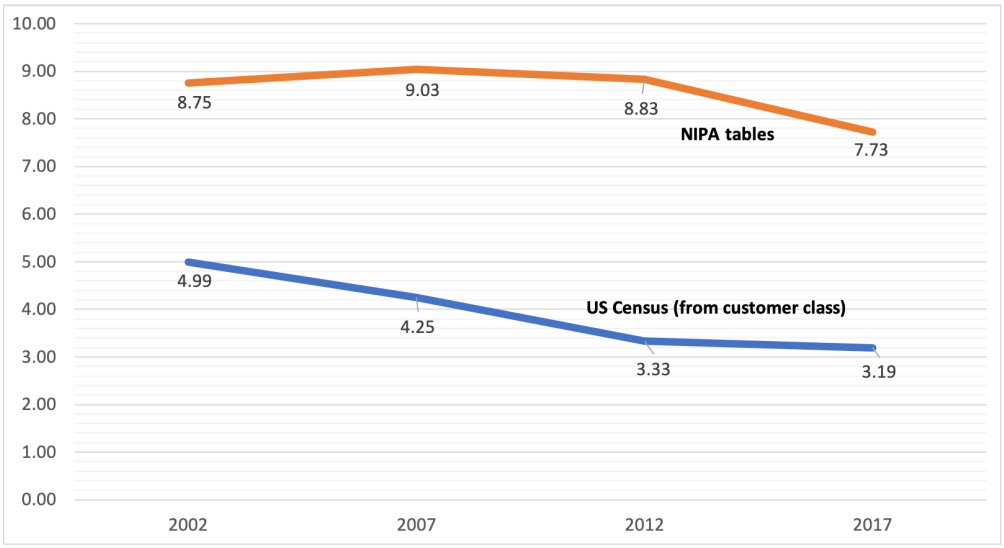
> The commodity-flow estimates may be improved by the incorporation of additional information, such as class-of-customer data. These data consist of information provided by sellers on the category of customer—usually households, business, and government—for their products or services. Class-of-customer data are collected in the Economic Census for several sectors but not for all industries, and the coverage varies from one census to another. Where class-of-customer data have been available, they have been used either to refine the commodity-flow estimates or to improve the estimate of the amount of a commodity sold to a particular group of consumers, such as the share of output sold to the various levels and functions of government.

[13] We use the information contained in the NIPA Chapter 9, "Government Consumption Expenditures and Gross Investment."

[14] To derive the figures on government purchases, we take the following steps:

1. For each four-digit industry group, we multiply the fraction of sales to the government as reported in the publicly tabulated statistics and the total sales.
2. We sum this statistic over all four-digit industry groups for a given year.

Figure 5. Government Purchases/GDP (%)



Notes: Figure 5 shows the evolution of procurement's total value in the US economy as measured in the NIPA tables versus US Census (using the customer class variable). Government purchases in the NIPA tables are calculated using Equation (1).

5 Contrasting EC Data with USAspending

This section compares the EC customer class data with USAspending, highlighting the strengths and limitations of each for studying government procurement. Understanding the differences across key dimensions—such as unit of analysis, sectoral coverage, and data frequency—helps researchers select the most appropriate dataset for their questions.

USAspending is a portal that has information about contracts awarded by the federal government. Data from the USAspending portal have been used widely in the literature to study government procurement. See footnote 1 for a list of prominent papers that have used data from USAspending to study a wide range of topics.

To frame our comparison between the EC and USAspending data, it is useful to highlight the key dimensions along which these datasets differ. We have identified the following areas in which USAspending data has advantages over the data from the EC:

- **Sectoral coverage**: The scope of USAspending is broader in terms of the sectors covered and includes firms in the Manufacturing sector and other sectors described in Table 3 (e.g., Transportation and Warehousing and Finance and Insurance sectors).[15]

[15]In addition to contracts, USAspending also includes other types of information, such as grants, cooperative agreements, and loans. Bruce, de Figueiredo and Silverman (2019) use USAspending data to study how the federal government arranges R&D spending with private sector firms. They focus their analysis on a comparison between the federal government's use of grants and cooperative agreements. Note that Arora, Belenzon and Suh (2022) use data from USAspending R&D contracts to construct changes in US federal contract R&D spending resulting from the end of the Cold War.

- **Buyer granularity**: USAspending has the advantage of having contract-level information that includes data about prime contractors and some data about first-tier subcontractors.[16] The data from the EC, on the other hand, would only cover establishments that sell directly to the government, such as prime contractors. The design of the EC forms suggests that subcontractors likely classify their customer as a private firm (the prime contractor) rather than a government entity, limiting the information related to indirect procurement.

- **Frequency and timing**: The EC takes place at five-year intervals, whereas USAspending provides exact contract award dates.

In contrast, the following are areas in which the data from the EC have an advantage over the data from USAspending:

- **Establishment level versus firm level**: EC data is available at the establishment level, whereas USAspending data is at the firm-level. This additional detail from the EC allows researchers to observe the sales of a firm to the government broken down by establishment.

- **Linkability to other datasets**: EC data can easily be linked to other Census datasets. Linking firm-level USAspending information to Census data would require name matching.

- **State and local coverage**: The EC provides information on sales to both federal and state/local governments, while USAspending focuses on federal contracts.

- **Long-term availability**: The EC has many years of data, especially in the Retail, Wholesale, and Construction sectors where continuous data extend back to at least 1972, allowing researchers to study long-term patterns. The USASpending portal was launched in 2007, although some of the sources cited in footnote 1 indicate that the portal contains data going back to the early 2000s.

[16] Prime contractors are the firms that have a direct contract with the federal government, and first-tier subcontractors are hired directly by a prime contractor to perform part of the work. Note that the reporting requirements include minimum values for awards for prime and first-tier subcontractors in order to show up in the USAspending dataset. According to the USAspending website:

> The Federal Funding Accountability and Transparency Act of 2006 (FFATA) was signed into law on September 26, 2006. The legislation required that federal contract, grant, loan, and other financial assistance awards of more than $25,000 be displayed on a publicly accessible and searchable website to give the American public access to information on how their tax dollars are being spent. In 2008, FFATA was amended by the Government Funding Transparency Act, which required prime recipients to report details on their first-tier sub-recipients for awards made as of October 1, 2010.

6 Alternative Sources for Class of Customer Data

The data from the class of customer question has many novel features relative to existing datasets that have been used to study government procurement. For example, the data can easily be linked to other Census data and is at the establishment level. On the other hand, not all sectors are covered by the class of customer question, and this coverage has changed over time. Thus, reviewing alternative datasets can help researchers identify complementary sources depending on the research question.

Below we discuss a few data sources that we found over the course of our research for this paper that contain class of customer information. We are not aware of any research study that has made use of these data sources to study government procurement. Note that while these datasets contain some information on customer types, they typically lack the establishment-level detail provided by the EC.

6.1 Annual Business Survey

The Annual Business Survey (ABS) is a firm-level survey from the Census that collects data about the demographic characteristics for businesses and business owners, as well as other topics such as innovation. The target population of the survey is nonfarm employer businesses filing the 941, 944, or 1120 tax forms.

Figure 6 shows the question that is asked in the 2017 ABS about a firm's class of customer. In the question, the survey asks firms to indicate a particular type of customer (e.g., U.S. Federal government) only if that type of customer accounts for at least 10 percent of revenue. Furthermore, firms are not asked about the fraction of sales that are accounted for by a particular customer class. Thus, because the ABS records whether a customer type accounts for at least 10 percent of sales, it is helpful to identify firms for which sales to the government account for at least 10 percent of their sales. On the other hand, it does not provide a measure of intensity (i.e., what fraction of the firm's sales are to the government). One useful feature of this data, however, is that it can easily be linked to other Census datasets. For example, researchers could complement the information from the ABS with the class of customer information from the EC.

Note that the ABS began collecting data for 2017. Before the ABS, the Survey of Business Owners (SBO) collected similar information about the class of customers of firms.[17] The SBO covers firms with paid employees and with no paid employees.

[17] For example, see question 66 on the 2012 SBO-1 form.

Figure 6. Class of Customer Question in the 2017 Annual Business Survey

TYPES OF CUSTOMERS

In 2017, which of the following types of customers accounted for 10% or more of this business's total sales of goods and/or services? **Select all that apply.**

□ U.S. Federal government

□ State and local government, including school districts, transportation authorities, etc.

□ Other businesses, including distributors of your product(s)

□ Other organizations (foreign governments, nonprofits, etc.)

□ Individuals

6.2 Quarterly Services Survey (QSS)

The Quarterly Services Survey (QSS) asks firms about the class of customers. The survey is on a quarterly frequency, covers selected service sectors, and is used by the Census to construct the QSS economic indicator. This information could be helpful for researchers who are interested in data about class of customer at a higher frequency, as it is issued quarterly. Whereas the EC provides five-year snapshots of establishment-level sales breakdowns, the QSS enables researchers to observe fluctuations in government demand for services at a quarterly frequency.

For example, in the third quarter of 2023, the QSS indicates that 21.3 percent of sales in the Management, Scientific, and Technical Consulting Services (5416) industry group was to the government. Note that the unit of analysis of the QSS is the firm, whereas for the EC it is the establishment. In this quarter, the class of customer information is available for the following NAICS codes: Information (51), Professional, Scientific, and Technical Services (54), and Administrative and Support and Waste Management and Remediation Services (56).[18] Figure 7 shows an example of the class of customer question asked in QSS.

[18]The third quarter 2023 release of QSS included the following NAICS codes: Utilities (22), Transportation and warehousing (48-49), Information (51), Credit Intermediation and Related Activities (522), Real Estate and Rental and Leasing (53), Professional, Scientific, and Technical Services (54), Administrative and Support and Waste Management and Remediation Services (56), Educational Services (61), Health Care and Social Assistance (62), Arts, Entertainment, and Recreation (71), Other Services (except Public Administration) (81).

Figure 7. Class of Customer Question in the Quarterly Services Survey (Form QSS-1A)

Question 6: CLASS OF CUSTOMER

What percentage of revenue reported in 5 was received from the following classes of customer in the

1. Household consumers and individual users
2. Business firms and not-for-profit organizations
3. Government (Federal, state, and local)

Sum of lines 1 through 3 should equal 100%.

6.3 Compustat

Public firms are required to report the name and sales of any customer that accounts for more than 10 percent of their revenue (including government).[19] Note that this data is at the firm level and is available for publicly-traded companies. This information is reported in Compustat and can be used by researchers. It can also be linked to Census data using the Compustat–SSEL Bridge created by Tello-Trillo and Streiff (2020).

7 Microdata

7.1 Establishment Responses to Class of Customer Question

To assess potential response biases in the EC microdata, we examine how response rates to the class of customer question vary by the establishment size. In particular, we want to study whether there are large differences in the response rates of establishments by the establishment size (measured by number of employees). This exercise helps us assess whether parts of the data should be used with caution. For example, a sharp decline in the response rates for a particular size category would mean that researchers should interpret results for size categories with low response coverage cautiously, as nonresponses may correlate with unobserved establishment traits.

[19] Tito (2019) has used this data to construct measures of US imports and exports based on whether the firm is identified as foreign. Note that she did not study government procurement using the Compustat data.

We thus estimate the following regression using data from the 2017 EC:

$$I_i = \sum_j \beta_j Q_{ij} + \gamma_{s(i)} + \varepsilon_{ij}, \tag{2}$$

where I_i is an indicator function with the value of 1 if establishment i gave information that the Census deemed usable for tabulation of class of customer statistics (i.e., a non-missing, valid response to the class of customer question, as determined by the Census Bureau) and 0 if not; Q_{ij} is an indicator variable with the value of 1 if establishment i belongs to size category j where j has a value of 1 if the establishment's March 12, 2017 employment is between 1 and 19 (small establishment), 2 if the employment is between 20 and 499 (medium establishment), and 3 if employment is more than 500 (large establishment); $\gamma_{s(i)}$ are six-digit industry fixed effects; and ε_i is an error term. We use data from 2017 from the following Census files: Census of Retail Trade (CRT), Census of Services (CSR), Census of Transportation, Communications, and Utilities (CUT), and Census of Wholesale Trade (CWH).[20]

7.2 Results

Table 9 reports the results of estimating equation (2). Column 1 reports the results with no industry fixed effects. Generally, we find small variation among the three different types of establishment sizes. For example, the response rate of small establishments on average is 65 percent, whereas medium establishments are 5 percentage points more likely to respond and large establishments are 6 percentage points less likely to respond. Note that, although these differences are statistically significant, we do not view them as being large enough to cause concerns when using the data by different establishment sizes. Researchers that are studying particular industries, however, would need to investigate further whether there is systematic variation by establishment size within that particular industry.

The second column estimates the same regression, except that we now include industry fixed effects. We find that the differences by establishment size become smaller: medium-sized establishments are 4 percentage points more likely to respond, and large establishments

[20] We use establishments that meet the following criteria:

- They were used in generating publicly tabulated data
- They were surveyed by the EC (i.e., were mailed a complete questionnaire)
- They had non-zero sales in 2017
- They had non-zero employment on March 12, 2017
- The establishments belong to an industry that was asked the class of customer question and where the class of customer question specifically included the government.

Table 9. Regression of Response Indicator on Establishment Size

	Response indicator (I_i)	
	(1)	(2)
Category 2 (20–499 employees)	0.0500***	0.0401***
	(0.0008)	(0.0009)
Category 3 (500+ employees)	−0.055***	−0.017***
	(0.0070)	(0.0069)
Constant	0.650***	0.653***
	(0.0004)	(0.0004)
NAICS six-digit fixed effects	No	Yes
Observations	1,774,000	1,774,000
R^2	0.0024	0.0864

Notes: Table 9 reports the results of the estimation of equation (2), where we regress an indicator variable with the value of 1 if the establishment provided information in the class of customer question that the Census deemed useful for generating publicly tabulated statistics on a dummy variable based on whether the establishment is between 1–19 employees, 20–499 employees, or 500+ employees. Column (1) reports the results from the regression estimation without the six-digit NAICS fixed effects. Column (2) reports the results with the six-digit NAICS fixed effects. The omitted category in the regression is Category 1 (1–19 employees), and we do not report estimated coefficients for the NAICS six-digit fixed effects. Note that the number of observations have been rounded to comply with Census Bureau disclosure rules. Standard errors are in parentheses.

*** $p < 0.01$.

are 2 percentage points less likely to respond. This finding indicates that once we control for industry, the differences in the response rate by establishment size become smaller.

Thus, we do not find evidence of large differences in response rate by establishment size.

8 Conclusion

The class of customer question from the EC gives researchers new and underexplored data that are very different from the sources that have been used up to now to study the effects of government procurement on businesses and the economy. This paper aims to serve as a guide for researchers who are interested in using this data to study government procurement or other topics of interest.

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Appendix: Additional Tables

Table A1. Industry groups with Customer Class Information by Census Year

1/3

NAIC4	Description	2002	2007	2012	2017
2211	Electric power generation, transmission and distribution	A	A	A	A
2212	Natural gas distribution	A	A	A	A
2213	Water, sewage and other systems	A	A	A	A
4231	Motor vehicle and motor vehicle parts and supplies merchant wholesalers	A	A	A	A
4232	Furniture and home furnishing merchant wholesalers	A	A	A	A
4233	Lumber and other construction materials merchant wholesalers	A	A	A	A
4234	Professional and commercial equipment and supplies merchant wholesalers	A	A	A	A
4235	Metal and mineral (except petroleum) merchant wholesalers	A	A	A	A
4236	Household appliances and electrical and electronic goods merchant wholesalers	A	A	A	A
4237	Hardware, plumbing and heating equipment and supplies merchant wholesalers	A	A	A	A
4238	Machinery, equipment, and supplies merchant wholesalers	A	A	A	A
4239	Miscellaneous durable goods merchant wholesalers	A	A	A	A
4241	Paper and paper product merchant wholesalers	A	A	A	A
4242	Drugs and druggists' sundries merchant wholesalers	A	A	A	A
4243	Apparel, piece goods, and notions merchant wholesalers	A	A	A	A
4244	Grocery and related product merchant wholesalers	A	A	A	A
4245	Farm product raw material merchant wholesalers	A	A	A	A
4246	Chemical and allied products merchant wholesalers	A	A	A	A
4247	Petroleum and petroleum products merchant wholesalers	A	A	A	A
4248	Beer, wine, and distilled alcoholic beverage merchant wholesalers	A	A	A	A
4249	Miscellaneous nondurable goods merchant wholesalers	A	A	A	A
4251	Wholesale electronic markets and agents and brokers	A	A	A	A
4411	Automobile dealers	A	A	A	A
4412	Other motor vehicle dealers	A	A	A	A
4413	Automotive parts, accessories, and tire stores	A	A	A	A
4421	Furniture stores	A	A	A	A
4422	Home furnishings stores	A	A	A	A
4431	Electronics and appliance stores	A	A	A	A
4441	Building material and supplies dealers	A	A	A	A
4442	Lawn and garden equipment and supplies stores	A	A	A	A
4451	Grocery stores	A	A	A	A
4452	Specialty food stores	A	A	A	A
4453	Beer, wine, and liquor stores	A	A	A	A
4461	Health and personal care stores	A	A	A	A
4471	Gasoline stations	A	A	A	A
4481	Clothing stores	A	A	A	A
4482	Shoe stores	A	A	A	A
4483	Jewelry, luggage, and leather goods stores	A	A	A	A
4511	Sporting goods, hobby, and musical instrument stores	A	A	A	A
4512	Book, periodical, and music stores	A	A	A	A
4521	Department stores	A	A	A	×
4522	Department stores	×	×	×	A

4523	General merchandise stores, including warehouse clubs and super-centers	×	×	×	A
4529	Other general merchandise stores	A	A	A	×

2/3

NAIC4	Description	2002	2007	2012	2017
4531	Florists	A	A	A	A
4532	Office supplies, stationery, and gift stores	A	A	A	A
4533	Used merchandise stores	A	A	A	A
4539	Other miscellaneous store retailers	A	A	A	A
4541	Electronic shopping and mail-order houses	A	A	A	A
4542	Vending machine operators	A	A	A	A
4543	Direct selling establishments	A	A	A	A
5111	Newspaper, periodical, book, and directory publishers	N	N	N	N
5112	Software publishers	N	N	N	N
5121	Motion picture and video industries	N	N	N	N
5122	Sound recording industries	N	N	N	N
5151	Radio and television broadcasting	N	N	N	N
5152	Cable and other subscription programming	N	N	N	N
5161	Internet publishing and broadcasting	N	×	×	×
5171	Wired telecommunications carriers	A	A	A	×
5172	Wireless telecommunications carriers (except satellite)	A	A	A	×
5173	Wired and wireless telecommunications carriers	A	×	×	A
5174	Satellite telecommunications	A	A	A	A
5175	Cable and other program distribution	A	×	×	×
5179	Other telecommunications	A	A	A	A
5181	Internet service providers	N	×	×	×
5182	Data processing, hosting, and related services	N	N	N	N
5191	Other information services	N	N	N	N
5311	Lessors of real estate	N	N	N	N
5312	Offices of real estate agents and brokers	N	N	N	N
5313	Activities related to real estate	N	N	N	N
5321	Automotive equipment rental and leasing	A	A	A	A
5322	Consumer goods rental	N	N	N	N
5323	General rental centers	A	A	A	A
5324	Commercial and industrial machinery and equipment rental and leasing	A	A	A	A
5331	Lessors of nonfinancial intangible assets (except copyrighted works)	N	N	N	N
5411	Legal services	N	F	F	F
5412	Accounting, tax preparation, bookkeeping, and payroll services	F	F	F	F
5413	Architectural, engineering, and related services	N	N	N	N
5414	Specialized design services	F	F	F	F
5415	Computer systems design and related services	F	F	F	F
5416	Management, scientific, and technical consulting services	F	N	N	N
5417	Scientific research and development services	F	F	F	F
5418	Advertising, public relations, and related services	F	F	F	F
5419	Other professional, scientific, and technical services	N	F	F	F
5611	Office administrative services	A	A	A	A
5612	Facilities support services	A	A	A	A
5613	Employment services	N	N	N	N
5614	Business support services	A	A	A	A
5615	Travel arrangement and reservation services	A	A	A	A
5616	Investigation and security services	A	A	A	A
5617	Services to buildings and dwellings	A	A	A	A
5619	Other support services	A	A	A	A
5621	Waste collection	A	A	A	A

5622	Waste treatment and disposal	A	A	A	A
5629	Remediation and other waste management services	A	A	A	A

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NAIC4	Description	2002	2007	2012	2017
7211	Traveler accommodation	N	A	A	A
7212	RV (recreational vehicle) parks and recreational camps	N	A	A	A
7213	Rooming and boarding houses	N	A	A	A
7221	Full-service restaurants	N	A	×	×
7222	Limited-service eating places	N	A	×	×
7223	Special food services	N	A	A	A
7224	Drinking places (alcoholic beverages)	N	A	A	A
7225	Restaurants and other eating places	×	×	A	A
8111	Automotive repair and maintenance	F	F	F	F
8112	Electronic and precision equipment repair and maintenance	F	F	F	F
8113	Commercial and industrial machinery and equipment (except automotive and electronic) repair and maintenance	F	F	F	F
8114	Personal and household goods repair and maintenance	F	F	F	F
8121	Personal care services	N	N	N	N
8122	Death care services	N	N	N	N
8123	Drycleaning and laundry services	N	N	N	N
8129	Other personal services	N	N	N	N
8132	Grantmaking and giving services	N	N	N	N
8133	Social advocacy organizations	N	N	N	N
8134	Civic and social organizations	N	N	N	N
8139	Business, professional, labor, political, and similar organizations	N	N	N	N

Notes: Table A1 shows the availability of customer class information for each four-digit NAICS category included in all versions of the Economic Census. Column 1 reports the four-digit NAICS code. Column 2 reports the name of the sector. Columns 3 through 6 indicate data availability as follows: "A" Information is available for all establishments; "F" Information is available only for establishments subject to federal tax; "N" Customer class information is not available; and "×" The category did not exist.

Table A2. Response Coverage for 2002, 2007, and 2012

NAICS2	Sector Name	Mean	Sd.
	2002		
22	Utilities	77.27	2.40
42	Wholesale trade	72.62	9.95
44	Retail trade	76.77	6.43
45	Retail trade	77.12	11.17
51	Information	51.55	14.76
53	Real estate and rental and leasing	80.07	5.26
54	Professional, scientific, and technical services	65.17	6.37
56	Administrative and support and waste management and remediation services	67.27	8.70
81	Other services (except public administration)	57.75	13.08
	Overall	70.93	11.89
	2007		
22	Utilities	84.37	4.31
42	Wholesale trade	76.43	4.22
44	Retail trade	80.45	6.30
45	Retail trade	83.07	9.80
51	Information	54.33	13.08
53	Real estate and rental and leasing	86.87	7.06
54	Professional, scientific, and technical services	69.91	7.28
56	Administrative and support and waste management and remediation services	65.40	6.32
72	Accommodation and food services	73.66	4.65
81	Other services (except public administration)	64.58	8.43
	Overall	75.02	10.25
	2012		
22	Utilities	82.40	4.22
42	Wholesale trade	72.94	6.73
44	Retail trade	81.51	6.58
45	Retail trade	72.12	16.47
51	Information	55.45	9.68
53	Real estate and rental and leasing	85.23	8.08
54	Professional, scientific, and technical services	70.19	7.41
56	Administrative and support and waste management and remediation services	67.53	3.28
72	Accommodation and food services	65.58	9.35
81	Other services (except public administration)	68.00	2.19
	Overall	72.77	10.71

Notes: Table A2 reports the response coverage for the years 2002, 2007, and 2012 by two-digit sectors for which the information is available. Response coverage is defined as the share of total sector sales contributed by establishments that provided usable responses to the question. Column 1 reports the NAICS two-digit sector. Column 2 reports the name of the sector. Column 3 reports the simple average of the response coverage for the four-digit NAICS codes within each sector and year. Column 4 is calculated in the same way but reports the standard deviation for that sector and year. The "Overall" row reports a simple average and standard deviation over all industries for which customer class information is available for that year. Note that publicly tabulated statistics on the response coverage rate is not available for the year 2017. For example, the publicly tabulated data reports that the response coverage of the class of customer inquiry for Management Consulting Services (54161) is between 50 and 60 percent, as discussed in the introduction. For that reason, we do not include data for 2017 in the table. In addition, response coverage information is not publicly available for the Construction sector, and thus we exclude this sector from the table.